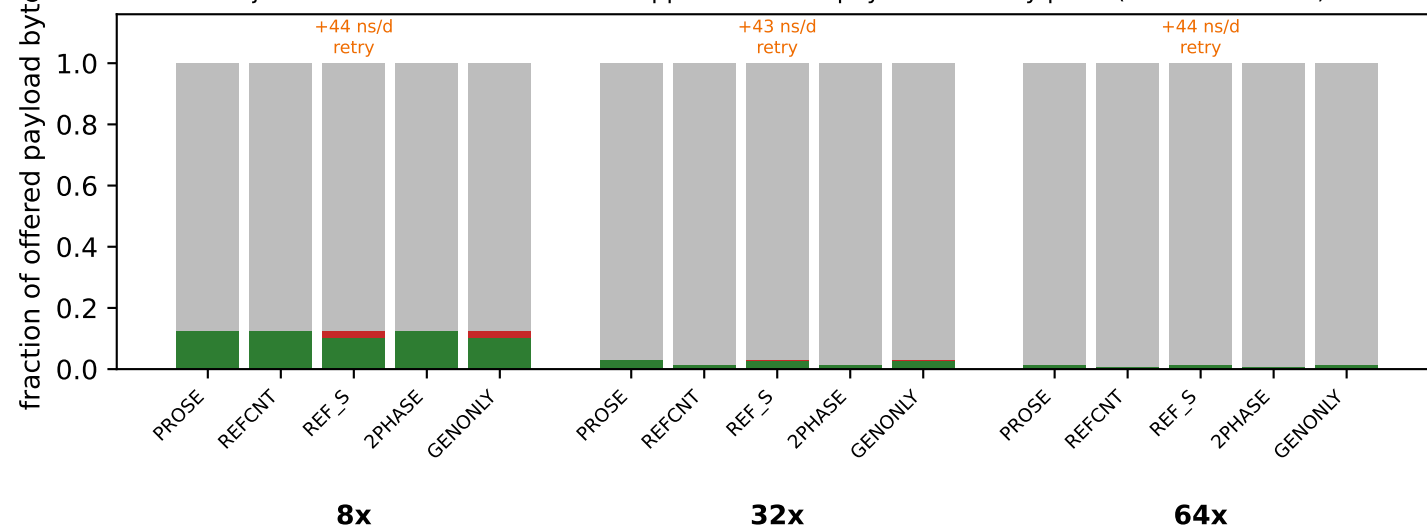


Cost of the commit-time gate: rejected descriptors pay metadata only; retries are a REFCNT_S-only tax; stale payload is the unsafe designs' tax

valid payload stale payload (RPE) rejected payload (never transferred)

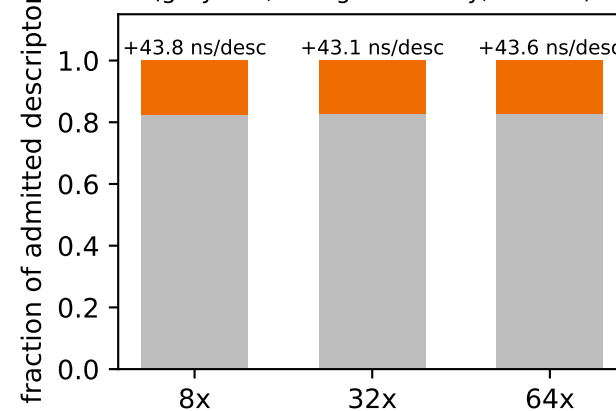
(a) offered-payload decomposition at 8/32/64x (5 main mechanisms)

reject-metadata overhead ≤ 969 ppm of offered payload at every point (too small to see)



(b) REFCNT_S retry distribution

(gray = 0, orange = 1 retry; 2+ = 0)



(c) reject vs stale-issue byte cost

(measured ratio $1.41e-03$; $META_B/CHUNK_PAYLOAD_B = 1/1024$)

